

A rational quadratic support formulation for the sharp gap

Exact finite realizability, a fixed energy matrix, and an explicit volume error

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Abstract

We replace the volume of a sample convex hull by a fixed quadratic objective on finitely many support values. The support points satisfy exact antipodal and diameter constraints, so every feasible finite array has a genuine constant-width completion. On an antipodal spherical triangulation, the piecewise linear homogeneous interpolant has a Dirichlet energy whose entire matrix is rational when the normals are rational. A boundary integration formula eliminates all angular integrals from that matrix. Uniform curvature bounds give an explicit quadratic-in-mesh error for its volume objective. The finite optimization is a concave quadratic minimization over a convex second-order-cone feasible set; it is not a convex optimization problem. Its exact global optima converge to the true constant-width minimum on shape-regular refinements. We give a rational sufficient certificate for an energy bound and exact checks of two finite matrices. No competitive global minimum is computed here, and no new numerical lower bound or sharpness claim is made.

1 The objective: keep actual support data and remove hull combinatorics

Let $K \subset \mathbb{R}^3$ have constant width one. Its support function satisfies $h(n) + h(-n) = 1$. Put

$$u = h - \frac{1}{2}, \quad u(-n) = -u(n), \quad \mathcal{Q}u = \nabla_{\mathbb{S}^2}^2 u + uI.$$

The curvature-radius tensor of K lies between zero and the identity:

$$-\frac{1}{2}I \preceq \mathcal{Q}u \preceq \frac{1}{2}I. \tag{1.1}$$

These statements have an almost-everywhere interpretation for $C^{1,1}$ support functions. The regularity follows, for example, from $K + (-K) = \overline{B}(0, 1)$: both support functions are convex summands of the unit-ball support function, giving bounded second derivatives on spherical charts. Standard support and mixed-volume facts are as in [1].

Define the quadratic form

$$E(v) = \int_{\mathbb{S}^2} (|\nabla v|^2 - 2v^2) \, d\sigma, \quad \mathcal{V}(v) = \frac{\pi}{6} - \frac{1}{4}E(v). \tag{1.2}$$

For the actual centered support u ,

$$\text{Vol}(K) = \mathcal{V}(u). \tag{1.3}$$

For completeness, the Codazzi property of $\mathcal{Q}h$ gives $\operatorname{div} \operatorname{cof}(\mathcal{Q}h) = 0$. Integration by parts then shows

$$S(K) = \int \det(\mathcal{Q}h) = \int \left(h^2 - \frac{1}{2} |\nabla h|^2 \right).$$

Together with Blaschke's identity $\operatorname{Vol}(K) = S(K)/2 - \pi/3$, and the vanishing mean of the odd function u , this proves equation (1.3). The eigenvalues of $-\Delta_{\mathbb{S}^2}$ are $l(l+1)$; hence $E(v) \geq 0$ for every odd $v \in H^1(\mathbb{S}^2)$, with equality precisely on linear functions $v(n) = a \cdot n$. This also follows by applying the mean-zero spherical Poincare inequality, whose equality space is the degree-one harmonics.

The proposal below has no independently assigned random curvature. Its finite variables are actual support points. The only error is the explicit interpolation error proved in section 4. The remaining difficulty is a global finite quadratic optimization, not an unmeasured geometric realizability error.

2 Exact finite support interpolation

Let $N = \{n_i\} \subset \mathbb{S}^2$ be finite and antipodally closed, with $n_{\bar{i}} = -n_i$. Prescribe points $z_i \in \mathbb{R}^3$ and optionally a fixed finite set Y of diameter at most one.

Lemma 2.1 (Diameter-one completion). *Every nonempty finite set of diameter at most one is contained in a convex body of constant width one.*

Proof. For a current finite set Z , let $C = \mathcal{B}(Z) = \bigcap_{z \in Z} \overline{B}(z, 1)$. The minimum containing ball of Z has radius at most $\sqrt{3}/8$: its center is a positive convex combination of at most four contact points, and the weighted squared-distance identity gives $R^2 \leq (1 - \sum \lambda_i^2)/2 \leq 3/8$. Thus C has a strictly feasible point.

For a unit n , choose x maximizing $n \cdot x$ on C . The active-normal rule gives $n = \sum a_i(x - z_i)$, where $a_i \geq 0$, $|x - z_i| = 1$, and $\sum a_i \geq 1$. If $w \in C$, then

$$(x - z_i) \cdot (w - x) \leq -\frac{1}{2} |w - x|^2,$$

so $|w - (x - n)|^2 \leq 1$. Consequently $Z \cup \{x, x - n\}$ still has diameter at most one: x lies in every old ball, $Z \subset C$, and $x - n \in \mathcal{B}(C)$. Adjoin such a pair successively in a countable dense sequence of directions. The sets stay in a fixed unit ball about an original point. The closed convex hull of their union has diameter at most one and has width one in every selected direction. Continuity of width in direction proves the claim. $\square$

Theorem 2.2 (Exact interpolation). *A width-one body containing Y has support point z_i in direction n_i for every i if and only if*

$$\boxed{z_i - z_{\bar{i}} = n_i, \quad |z_i - z_j| \leq 1, \quad |z_i - y| \leq 1 \quad (y \in Y),} \quad (2.1)$$

with the already assumed diameter constraints within Y .

Proof. Opposite support points of a width-one body have scalar separation one in their normal direction and Euclidean distance at most one. Their difference is therefore exactly that unit normal, proving necessity. Conversely apply theorem 2.1 to the finite set in equation (2.1). If x belongs to the resulting body, then

$$n_i \cdot x \leq n_i \cdot z_i + |x - z_i| \leq n_i \cdot z_{\bar{i}} + 1 = n_i \cdot z_i.$$

As z_i is in the body, it is the prescribed support point. $\square$

The feasible set is convex. Every constraint in equation (2.1) is affine or a second-order-cone constraint. Its support values and centered values are

$$h_i = n_i \cdot z_i, \quad u_i = h_i - \frac{1}{2}, \quad u_{\bar{i}} = -u_i.$$

For the unrestricted problem, include $\pm e_3$ in N and impose $z_{e_3} = e_3/2$, $z_{-e_3} = -e_3/2$. Every body admits this translation, and it makes the finite feasible set compact. With a nonempty fixed seed, compactness follows instead from a unit ball about one seed point.

3 A fixed rational energy matrix

Let $\mathcal{T}$ be an antipodally symmetric triangular surface with vertices $n_i \in \mathbb{S}^2$, such that radial projection maps its faces bijectively, up to shared edges, onto a triangulation of $\mathbb{S}^2$. For example, triangulate the facets of an origin-containing inscribed convex polyhedron, consistently at antipodes. Every face lies in an open hemisphere.

For an oriented face $\sigma = (n_0, n_1, n_2)$, order its vertices so that $\det N_\sigma > 0$, where N_σ has rows n_0^T, n_1^T, n_2^T . Let $\Omega_\sigma \subset \mathbb{S}^2$ be its radial image. Define

$$B_\sigma = N_\sigma^{-1}(u_0, u_1, u_2)^T, \quad u_{\mathcal{T}}(n) = B_\sigma \cdot n \quad (n \in \Omega_\sigma).$$

This interpolant is continuous: on a shared edge, the difference of the two linear forms vanishes on the plane spanned by its two endpoints. It is piecewise smooth, lies in $H^1(\mathbb{S}^2)$, and is odd. It need not itself be a convex support function; exact realizability is supplied by theorem 2.2, not by imposing convexity on this interpolant.

Theorem 3.1 (Rational boundary formula). *For every face,*

$$E_\sigma(u_{\mathcal{T}}) = - \sum_{(i,j)=(0,1),(1,2),(2,0)} \frac{[B_\sigma \cdot (n_i \times n_j)][B_\sigma \cdot (n_i + n_j)]}{1 + n_i \cdot n_j}. \quad (3.1)$$

In particular, if all n_i are rational, there is an explicitly computable rational symmetric matrix $E_{\mathcal{T}}$ such that

$$E(u_{\mathcal{T}}) = \mathbf{u}^T E_{\mathcal{T}} \mathbf{u}. \quad (3.2)$$

On the antipodally odd nodal subspace this matrix is positive semidefinite. Its kernel is exactly the three-dimensional space $u_i = a \cdot n_i$.

Proof. A linear spherical function $\ell(n) = B \cdot n$ satisfies $\Delta \ell = -2\ell$. Green's identity on Ω_σ gives

$$\int_{\Omega_\sigma} (|\nabla \ell|^2 - 2\ell^2) = \int_{\partial\Omega_\sigma} \ell \partial_\nu \ell \, ds.$$

The outward spherical conormal on the oriented great-circle edge from n_i to n_j is the constant vector $\nu = -(n_i \times n_j)/|n_i \times n_j|$. If the edge length is $\theta \in (0, \pi)$, direct integration on its great circle yields

$$\int_{\text{edge}} n \, ds = (n_i + n_j) \tan(\theta/2), \quad \frac{\tan(\theta/2)}{|n_i \times n_j|} = \frac{1}{1 + n_i \cdot n_j}.$$

Also $\partial_\nu \ell = B \cdot \nu$. Substitution proves equation (3.1). Everything in this expression is rational in the vertices and quadratic in their nodal values. Summing faces gives equation (3.2). Positivity and the kernel follow from the spherical Poincare inequality, since the assembled interpolant is odd. A zero-energy interpolant is a global linear spherical function, and conversely every such function is interpolated exactly. $\square$

Here is an explicit matrix assembly formula. Put

$$T_\sigma = -\frac{1}{2} \sum_{(i,j)} \frac{(n_i \times n_j)(n_i + n_j)^T + (n_i + n_j)(n_i \times n_j)^T}{1 + n_i \cdot n_j}.$$

The local matrix on the three nodal values is $N_\sigma^{-T} T_\sigma N_\sigma^{-1}$. Add it to the corresponding three-by-three block of the global matrix. Only after assembly restrict to one value per antipodal pair. A single local matrix need not be positive semidefinite.

The cancellation of all angular integrals is useful: the program's quadratic coefficients are exact fractions, not numerical surface integrals. In contrast, direct convex-hull volume changes its algebraic expression when the sample hull changes facets. Here the triangulation and quadratic matrix are fixed before the unknown support data are chosen.

4 An explicit volume error independent of the unknown body

For each flat face, define

$$c_\sigma = \text{dist}(0, \text{aff } \sigma) > 0, \quad d_\sigma = \max_{i,j \in \sigma} |n_i - n_j|,$$

$$\epsilon_\sigma = \frac{1 - c_\sigma^2}{4c_\sigma^2}, \quad b_\sigma = \frac{\sqrt{3} d_\sigma^2}{4c_\sigma \sigma_{\min}(N_\sigma)}, \quad \epsilon = \max_\sigma \epsilon_\sigma, \quad b = \max_\sigma b_\sigma. \quad (4.1)$$

Here $\sigma_{\min}$ denotes the smallest singular value. No unknown support function occurs in these constants.

Lemma 4.1 (Uniform interpolation estimates). *For any actual width-one centered support u interpolating the nodal data,*

$$\|u_{\mathcal{T}} - u\|_\infty \leq \epsilon, \quad |\nabla(u_{\mathcal{T}} - u)| \leq b \quad \text{almost everywhere.} \quad (4.2)$$

Proof. Extend u homogeneously by $\hat{u}(x) = |x|u(x/|x|)$. Its Euclidean Hessian has radial eigenvalue zero and tangential block $|x|^{-1} \mathcal{Q}u(x/|x|)$. By equation (1.1),

$$\|D^2 \hat{u}(x)\| \leq \frac{1}{2|x|}.$$

On a flat face, $|x| \geq c_\sigma$, and the same holds on every segment in that face. Hence $\nabla \hat{u}$ is Lipschitz there with constant $1/(2c_\sigma)$.

Write $x = \sum \lambda_i n_i$, with $\lambda_i \geq 0$, $\sum \lambda_i = 1$, and $r = |x|$. The affine interpolation remainder is bounded by

$$\left| \sum_i \lambda_i u_i - \hat{u}(x) \right| \leq \frac{1}{4c_\sigma} \sum_i \lambda_i |n_i - x|^2 = \frac{1 - r^2}{4c_\sigma}.$$

Since $u_{\mathcal{T}}(x/r) = r^{-1} \sum \lambda_i u_i$, division by $r \geq c_\sigma$ proves the value estimate.

For the gradient, let $a = \nabla \hat{u}(x)$. Euler's identity gives $a \cdot x = \hat{u}(x)$. Taylor's formula at each vertex thus gives

$$(B_\sigma - a) \cdot n_i = u_i - \hat{u}(x) - a \cdot (n_i - x), \quad |(B_\sigma - a) \cdot n_i| \leq \frac{d_\sigma^2}{4c_\sigma}.$$

Inverting N_σ yields $|B_\sigma - a| \leq b_\sigma$. Spherical gradients are the orthogonal projections of these homogeneous Euclidean gradients, so their difference satisfies the same bound. $\square$

A purely algebraic replacement for the singular-value bound is

$$b_\sigma^2 \leq \frac{9d_\sigma^6}{8c_\sigma^2(\det N_\sigma)^2}. \quad (4.3)$$

Indeed $\sigma_1(N_\sigma) \leq \sqrt{3}$. Subtracting the rank-one matrix with all rows n_0^T gives $\sigma_2(N_\sigma) \leq \sqrt{2}d_\sigma$. Their product and the determinant identity then give $\sigma_{\min} \geq |\det N_\sigma|/(\sqrt{6}d_\sigma)$. Both c_σ^2 and the right side of equation (4.3) are rational when the vertices are rational.

Theorem 4.2 (Uniform quadratic-objective error). *Every genuine completion of an exact finite array satisfies*

$$\boxed{\mathcal{V}(u_{\mathcal{T}}) - 2\pi\epsilon \leq \text{Vol}(K) \leq \mathcal{V}(u_{\mathcal{T}}) + 2\pi\epsilon + \pi b^2.} \quad (4.4)$$

Proof. Let $w = u_{\mathcal{T}} - u$, which is odd, and let $L = -\Delta - 2$. By equation (1.1), $Lu = -\text{tr } \mathcal{Q}u$ lies in $[-1, 1]$ almost everywhere. Weak integration by parts is legitimate for $u \in C^{1,1}$ and $w \in H^1$. Expanding the energy gives the exact identity

$$\text{Vol}(K) - \mathcal{V}(u_{\mathcal{T}}) = \frac{1}{2} \int (Lu)w + \frac{1}{4}E(w).$$

The first term has absolute value at most $2\pi\epsilon$. Oddness gives $E(w) \geq 0$, while $E(w) \leq \int |\nabla w|^2 \leq 4\pi b^2$. This proves both bounds. Notice that the lower bound does not require shape regularity or the gradient-error constant b . $\square$

On a shape-regular sequence of meshes with maximum edge length $h \rightarrow 0$, $c_\sigma \rightarrow 1$ uniformly, $1 - c_\sigma^2 = O(h^2)$, and $|\det N_\sigma|$ is comparable to d_σ^2 . Consequently $\epsilon + b^2 = O(h^2)$. The needed geometric condition can be checked directly from equations (4.1) and (4.3); there is no appeal to regularity of an unknown minimizing body. Antipodally symmetric rational perturbations of shape-regular refinements preserve these bounds if their displacement is sufficiently smaller than the mesh size.

5 The finite optimization and a rational certificate target

Let $\mathcal{Z}_{\mathcal{T}}$ be the compact feasible set in equation (2.1), with the diametral normalization when there is no fixed seed. Define

$$Q_{\mathcal{T}} = \max_{z \in \mathcal{Z}_{\mathcal{T}}} \mathbf{u}(z)^T \mathbf{E}_{\mathcal{T}} \mathbf{u}(z), \quad q_{\mathcal{T}} = \frac{\pi}{6} - \frac{1}{4}Q_{\mathcal{T}}.$$

The finite objective is convex in the support values when written as the energy to be maximized. Equivalently its volume form is a concave quadratic to be minimized. Either optimization is generally nonconvex.

Theorem 5.1 (A finite quadratic enclosure of the true minimum). *Let V_* be the unrestricted width-one minimum, or the minimum in the class containing the prescribed seed. Then*

$$\boxed{q_{\mathcal{T}} - 2\pi\epsilon \leq V_* \leq q_{\mathcal{T}} + 2\pi\epsilon + \pi b^2.} \quad (5.1)$$

In particular, if $\epsilon + b^2 \rightarrow 0$, then $q_{\mathcal{T}} \rightarrow V_$.*

Proof. Every body produces a feasible finite array. Its interpolated quadratic value is at least $q_{\mathcal{T}}$, so the lower bound follows from theorem 4.2. Conversely, a minimizing finite array exists by compactness and continuity. It has an actual completion by theorem 2.2. The upper bound in theorem 4.2, applied to that completion, proves the other inequality. $\square$

The finite values $q_{\mathcal{T}}$ are not claimed to be monotone under refinement. Monotone lower and upper sequences can be formed by taking, respectively, running maxima of the certified lower endpoints and running minima of the certified upper endpoints in equation (5.1).

For a practical sufficient certificate, eliminate all affine equations and write the remaining support-point variables as a vector x . The diameter constraints take the form

$$g_e(x) = 1 - |D_e x + d_e|^2 \geq 0, \quad \mathbf{u} = Ux + v.$$

For rational normals and a rational fixed seed, all these coefficients and $\mathbf{E}_{\mathcal{T}}$ are rational. Any exact identity

$$B - (Ux + v)^T \mathbf{E}_{\mathcal{T}} (Ux + v) = \begin{pmatrix} 1 \\ x \end{pmatrix}^T Z \begin{pmatrix} 1 \\ x \end{pmatrix} + \sum_e \lambda_e g_e(x), \quad Z \succeq 0, \quad \lambda_e \geq 0, \quad (5.2)$$

proves $Q_{\mathcal{T}} \leq B$. It therefore implies

$$V_* \geq \frac{\pi}{6} - \frac{B}{4} - 2\pi\epsilon.$$

Coefficient equality and a rational positive-semidefinite factorization of Z are finite verification tasks. This is only a sufficient certificate; we do not assert that constant multipliers λ_e always attain the exact finite maximum. Branch-and-bound or stronger polynomial certificates may be necessary. An optimizer's feasible array supplies no upper bound on $Q_{\mathcal{T}}$.

To reach the sharp Meissner value by this formulation, it would suffice to prove, for meshes with $\epsilon \rightarrow 0$,

$$Q_{\mathcal{T}} \leq E_M + \delta_{\mathcal{T}}, \quad \delta_{\mathcal{T}} \rightarrow 0, \quad E_M = 4 \left(\frac{\pi}{6} - V_M \right) = \pi\sqrt{3} \arccos \frac{1}{3} - 2\pi.$$

No such family of upper certificates is produced here. The new result makes the required objective a fixed rational quadratic form while preserving finite realizability; it does not solve its global maximization.

6 Exact algebraic checks and numerical status

The accompanying standard-library script constructs two rational normal sets by stereographic grids of orders one and two, with duplicates removed. It enumerates all supporting facets exactly, triangulates them antipodally, and verifies oriented edge cancellation, vertex use, Euler's identity, and positive radial orientation. For each triangulation it constructs the matrix in theorem 3.1 using fractions only, restricts to odd values, and performs an exact semidefinite elimination. It also checks that all three translation vectors lie in its kernel.

For a nonspherical feasible test body, take

$$h(n) = \frac{1}{2} + \frac{1}{10} n_1 n_2 n_3.$$

If $P(X) = X_1X_2X_3$, then $\mathcal{Q}(P|_{\mathbb{S}^2}) = D^2P|_{T\mathbb{S}^2} - 2PI$. On the sphere, $\|D^2P\| \leq 2$ and $|P| \leq 1/5$, so the perturbation's curvature norm is at most $6/25 < 1/2$. The displayed h is therefore a width-one support function. Its support points at rational normals are rational:

$$z(n) = \left(\frac{1}{2} - \frac{1}{5}P(n)\right)n + \frac{1}{10}\nabla P(n).$$

The program checks all finite diameter inequalities and antipodal identities for these data. Since $P|_{\mathbb{S}^2}$ has harmonic degree three and $\int P^2 d\sigma = 4\pi/105$, its exact volume is

$$\frac{\pi}{6} - \frac{\pi}{1050}.$$

This is a test body, not a minimum. Its computed finite energy is a check of the rational matrix construction; the rigorous error estimates above are proved analytically and can be coarse on these two small meshes.

The executed checks give:

Grid	Vertices	Triangles	Odd variables	Energy rank	Nullity
1	14	24	7	4	3
2	38	72	19	16	3

The two exact test energies are $16/2025$ and $368/54675$. They are values for the displayed test body, not maxima over feasible arrays. All 91 and 703 pairwise diameter inequalities, respectively, pass. The complete rational matrices and triangulations are supplied as JSON data.

Run `python check_rational_energy.py`. The script prints and writes its exact counts, rational energies, kernel dimensions, and geometric error constants. It does not solve a global quadratic optimization, certify a new volume lower bound, or verify the sharp inequality. The universal value 0.4144 is established in the separate main manuscript and its fully rerun certificate; the sharp gap is still open in this continuation.

References

- [1] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, second expanded edition, Encyclopedia of Mathematics and its Applications 151, Cambridge University Press. <https://doi.org/10.1017/CBO9781139003858>.
- [2] R. Hynd, *On extreme constant width bodies in $\mathbb{R}^3$* , arXiv:2501.16940. Background on support-function constraints; the finite interpolation and error estimates used here are proved in the text.